

Predicting The Sports Market Business Using ML

submitted in fulfilment of the requirements for Major Project-1/2 Sem-VIII

Bachelor of Engineering

By

Mr. Paras Adkurkar (Roll No. 01)

Mr. Abhishek Barote (Roll No. 11)

Mr. Prathmesh Bhoir (Roll No. 16)

Mr. Aniket Chavan (Roll No. 25)

Under the guidance of

Prof. Sunil Katkar



Department of Computer Engineering
Vidyavardhini's College of Engineering & Technology, Vasai (W)
(NAAC and NBA Accredited)
(Affiliated with University of Mumbai)

(2025-2026)

CERTIFICATE

This is to certify that the project entitled “**Predicting The Sports Market Business Using ML**” is a bonafide work of “**Paras Adkurkar (Roll No. 1), Abhishek Barote (Roll No. 11), Prathmesh Bhoir (Roll No. 16) and Aniket Chavan (Roll No. 25)**” submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of “**Bachelor of Engineering**” in “**Computer Engineering**”.

Prof. Sunil Katkar
Guide

Dr. Megha Trivedi
Head of Department

Dr. Rakesh Himte
Principal

Project Report Approval for B.E.

This project report entitled “**Predicting The Sports Market Business Using ML**” by “**Paras Adkurkar (Roll No. 1), Abhishek Barote (Roll No. 11), Prathmesh Bhoir (Roll No. 16)** and **Aniket Chavan (Roll No. 25)**” is approved for the degree of “**Bachelor of Engineering**” in “**Computer Engineering**”.

Examiners

1. _____

2. _____

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Paras Adkurkar (01)

Abhishek Barote (11)

Prathmesh Bhoir (16)

Aniket Chavan (25)

Date:

Acknowledgements

We would like to express our heartfelt gratitude to Vidyavardhini's College of Engineering and Technology for providing the opportunity to work on our project, "Predicting The Sports Market Business Using ML." We sincerely thank our Principal, Dr. Rakesh Himte, and our Head of Department, Dr. Megha Trivedi, for their continuous support and encouragement. Our deepest appreciation goes to our Project Guide, Prof. Sunil Katkar, for her mentorship and guidance throughout the project. We also acknowledge the collaborative efforts of our team members, whose hard work enabled us to complete the project within the given timeframe. This project has greatly enhanced our knowledge in Machine Learning, providing valuable hands-on experience and real-world application.

Paras Adkurkar (01)

Abhishek Barote (11)

Prathmesh Bhoir (16)

Aniket Chavan (25)

Date:

Abstract

The sports industry has evolved into a highly dynamic, competitive, and data-intensive sector, where accurate forecasting of market trends plays a critical role in strategic planning, revenue optimization, and enhancing fan engagement. This project, “Predicting The Sports Market Business Using ML” aims to leverage historical and real-time data, including team performance metrics, match outcomes, ticket sales, merchandise revenue, social media engagement, fan sentiments, sponsorship activities, and broadcast viewership statistics, to predict key business outcomes and trends in the sports market. Advanced machine learning algorithms, such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks, are applied to capture non-linear patterns, temporal dependencies, and interactions among various influencing factors. Feature engineering techniques, including moving averages, trend indicators, sentiment analysis, and correlation analysis, are utilized to improve predictive performance and model accuracy. The project also emphasizes the development of an interactive visualization dashboard, which allows stakeholders to interpret the predicted outcomes, monitor trends, and make data-driven decisions related to ticket pricing, marketing campaigns, merchandise inventory management, sponsorship planning, and broadcast scheduling. By integrating machine learning with business intelligence, this project not only demonstrates the potential of predictive analytics in the sports domain but also highlights the practical significance of data-driven strategies in maximizing operational efficiency, improving financial performance, and gaining a competitive edge. The study further provides insights into emerging patterns in fan behavior, market dynamics, and revenue streams, enabling sports organizations to plan proactively, reduce financial risks, and enhance overall strategic decision-making in a rapidly evolving global sports industry.

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Chapter 1

Introduction

1.1 Problem Statement

The sports industry produces vast amounts of complex and ever-changing data from diverse sources such as ticket sales, merchandise purchases, sponsorship agreements, broadcast viewership figures, and social media engagement. However, despite having access to this information, many sports organizations struggle to accurately forecast market trends, revenue streams, and fan behavior. Conventional forecasting techniques often depend on basic statistical models that fail to capture the intricate, non-linear relationships and time-based dependencies between multiple factors affecting the sports business. Consequently, organizations encounter difficulties in optimizing ticket pricing, designing marketing campaigns, managing merchandise stock, and evaluating sponsorship returns — leading to inefficiencies, lost opportunities, and potential revenue declines.

To overcome these limitations, a robust, data-driven solution leveraging advanced machine learning methods is essential. By utilizing algorithms such as Random Forest, XGBoost, and LSTM networks, the system can uncover hidden patterns, forecast trends, and generate actionable insights for informed strategic decisions. This project focuses on building such a predictive framework, incorporating feature engineering, sentiment analysis, and interactive visualizations to assist decision-makers in sports organizations. The ultimate objective is to enhance operational efficiency, boost financial outcomes, and strengthen competitiveness through proactive planning, risk management, and optimized revenue strategies in the fast-evolving sports landscape.

1.2 Project Aim and Objective

The primary objective of this project is to develop an intelligent, machine learning-based predictive framework designed to analyze and forecast critical business trends within the sports industry. With the ever-growing influx of data from diverse sources such as ticket sales, merchandise transactions, sponsorship deals, broadcast viewership, and social media activity, sports organizations face increasing pressure to make data-driven strategic decisions. This project aims to harness both historical and real-time data to accurately predict revenue streams, fan

engagement patterns, and sponsorship returns. By uncovering hidden correlations and complex relationships between multiple factors influencing the sports business, the system will empower decision-makers to anticipate future market dynamics, optimize pricing strategies, plan marketing initiatives, and manage inventory more effectively. Ultimately, the project strives to provide a solid analytical foundation for improving organizational efficiency, maximizing profits, and maintaining a competitive edge in the fast-evolving sports market.

To achieve these goals, the project involves several key components, including comprehensive data collection and preprocessing from multiple sources such as team performance records, ticketing databases, sponsorship portfolios, and social media interactions. Advanced feature engineering, trend detection, and sentiment analysis techniques will be applied to refine the input data and enhance the accuracy of predictions. Various machine learning models—such as Random Forest, XGBoost, and LSTM networks—will be implemented and compared to identify the most effective approach for forecasting business outcomes. Additionally, the framework is designed to be scalable, cost-efficient, and adaptable, ensuring it can handle large datasets and real-time data streams without significant computational overhead. Through interactive visualizations and intuitive dashboards, the system will deliver actionable insights to support data-driven decision-making and long-term strategic planning for sports organizations.

1.3 Project Scope

The scope of this project involves building a comprehensive machine learning-based framework capable of accurately predicting essential business trends in the sports industry, including ticket sales, merchandise revenue, and sponsorship returns. The project entails collecting and preprocessing both historical and real-time data from various sources such as team performance statistics, match results, social media engagement metrics, broadcast viewership data, and sales transactions. To effectively model complex relationships and time-dependent patterns, advanced machine learning techniques such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks are employed. These algorithms are integrated with processes like feature engineering, trend analysis, and sentiment evaluation to enhance model precision and predictive capability. Moreover, the project includes the development of an interactive dashboard that provides clear visualizations of key insights and forecasts, enabling sports organizations to make informed decisions regarding ticket pricing, merchandise inventory management, marketing campaigns, and sponsorship evaluation.

The project's scope is primarily limited to analyzing and forecasting revenue-related aspects within the sports industry, rather than predicting individual player performance or match-specific outcomes. Although the current framework centers on major revenue streams—ticketing, merchandising, and sponsorship—it can be extended in future iterations to cover additional applications, such as audience engagement, marketing optimization, or overall financial planning. By combining predictive analytics with visualization tools, this project aims to deliver actionable insights that empower sports organizations to improve operational efficiency, manage risks effectively, and make strategic, data-driven business decisions in a competitive and continuously evolving market environment.

1.4 Motivation and Significance

1.4.1 Motivation

The sports industry has experienced remarkable growth in recent years, evolving into one of the most data-driven and competitive sectors globally. Massive amounts of data are continuously generated through various channels, including ticket sales, merchandise purchases, sponsorship agreements, broadcast ratings, and social media interactions. Despite having access to such diverse data sources, many sports organizations still struggle to make precise predictions about market trends, revenue streams, and fan engagement. Traditional forecasting models often rely on limited statistical approaches that fail to capture the complex, non-linear relationships and time-dependent patterns inherent in the data. As a result, organizations face difficulties in making informed decisions related to marketing, pricing, and inventory management—leading to inefficiencies, missed opportunities, and potential revenue losses.

The advent of advanced analytics and machine learning technologies provides a promising solution to these challenges. Machine learning algorithms such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks are capable of processing large datasets, uncovering hidden correlations, and generating highly accurate forecasts. Leveraging these algorithms enables sports organizations to predict key business indicators—such as ticket sales, merchandise demand, and sponsorship returns—with greater reliability. This project is driven by the vision of utilizing machine learning to enhance data-driven decision-making in the sports industry. By integrating predictive modeling with interactive visualization dashboards, the proposed system aims to deliver actionable insights that support proactive planning, efficient resource

management, and targeted marketing strategies. Ultimately, the project seeks to empower sports organizations to optimize operations, maximize profitability, and strengthen their competitive position in an ever-evolving and dynamic market landscape.

1.4.2 Significance

The project titled Predicting The Sports Market Business Using ML plays a vital role in today's increasingly data-centric sports industry. As the global sports market continues to expand, enormous volumes of data are being generated daily from diverse sources such as ticket sales, player statistics, sponsorships, fan interactions, and social media activities. Traditional analytical techniques often fall short in processing and interpreting such complex and large-scale data, limiting their ability to provide actionable insights. This project addresses these challenges by leveraging advanced machine learning algorithms to analyze and forecast market trends, audience preferences, and revenue patterns with higher accuracy. Through predictive analytics, it empowers decision-makers in the sports domain to develop informed strategies, optimize operations, and enhance overall business performance.

In addition to its analytical capabilities, the project significantly contributes to improving marketing strategies, optimizing ticket pricing, forecasting merchandise demand, and evaluating sponsorship returns. It aids sports organizations, analysts, and investors in identifying lucrative opportunities, minimizing financial risks, and making data-driven decisions that strengthen their market presence. Beyond financial optimization, the project's broader impact lies in promoting sustainable growth through efficient resource allocation and long-term strategic planning. By integrating artificial intelligence and machine learning into sports analytics, the initiative encourages innovation and paves the way for further research and technological advancements in this field. As the sports industry continues to embrace digital transformation, such AI-driven systems will become essential for maintaining competitiveness, improving transparency, and ensuring profitability across the global sports ecosystem.

Chapter 2

Literature Review

2.1 Existing system

Existing systems for forecasting sports market trends have traditionally been reliant on human expertise and descriptive analytics, which primarily focus on historical data to explain past performance rather than predict future outcomes. For decades, sports organizations have based critical market decisions—such as merchandise production, ticket pricing, and sponsorship valuation—on the intuition of seasoned executives and manual analysis of past sales reports. Ljubica and Seric (2018) [1] detail these traditional market research methods, which include fan surveys, focus groups, and simple environmental monitoring, all of which are time-consuming, expensive, and offer only a static snapshot of a highly dynamic market. These methods are inherently reactive, struggling to keep pace with rapid shifts in fan sentiment, player popularity, and media narratives.

The first significant evolution beyond manual analysis was the adoption of Business Intelligence (BI) platforms and basic statistical modeling. As detailed by Miller (2011) [2], the "Moneyball" era ushered in an acceptance of data, though its focus was almost exclusively on optimizing on-field player performance, not on commercial market dynamics. In the existing commercial landscape, organizations employ BI tools like Tableau or SAP Sports One to create dashboards that visualize historical key performance indicators (KPIs), such as ticket sales, concession revenues, and fan demographics (Grand View Research, 2024) [3]. While valuable for descriptive analysis, these systems lack predictive power. They can effectively report *what* merchandise sold after a championship win but cannot accurately forecast the *demand* for a specific rookie's jersey *before* they play a single professional game.

The application of machine learning (ML) in the current sports analytics market, while growing, remains highly fragmented and siloed. Existing ML solutions typically address narrow, isolated problems rather than providing a holistic market forecast. For instance, researchers have employed classical time-series models like ARIMA (Autoregressive Integrated Moving Average) to forecast seasonal ticket attendance based on historical patterns and team schedules (Baker et al., 2016) [4]. Separately, other systems may use basic Natural Language Processing (NLP) for

sentiment analysis, gauging fan reactions on social media platforms like Twitter to provide a rudimentary measure of "fan engagement" (Achen et al., 2016) [5]. These tools operate in isolation, failing to connect how social media sentiment might directly influence merchandise sales or sponsorship value.

More advanced, yet still limited, systems utilize supervised learning models to predict specific, binary outcomes. For example, some organizations use decision trees or logistic regression to forecast the likelihood of a season ticket holder renewing their membership, as described by Wiggins et al. (2018) [6]. These models are typically trained on structured datasets, such as fan transaction history, tenure, and demographic data. However, they fail to incorporate the vast, unstructured, and real-time data streams that heavily influence market trends, such as breaking news, player trade rumors, video highlights, or the sudden emergence of a cultural moment (e.g., a viral player celebration). The models are specific, predictive for one vertical, but not integrated into a comprehensive market-level forecast.

Comprehensive surveys of the sports analytics market, such as those by MarketsandMarkets (2024) [7] and SNS Insider (2024) [8], highlight a significant gap in the existing technological landscape. The market is populated by numerous point solutions for *performance analytics* (e.g., Catapult, Stats Perform), *fan relationship management* (e.g., EngageRM), and *video analysis* (e.g., Hudl), but it lacks a unified platform that integrates these disparate data sources for *commercial forecasting*. The primary challenge identified is the "fragmented data ecosystems" [7] that prevent organizations from modeling the complex, interdependent relationships between on-field performance, player popularity, media coverage, and fan purchasing behavior.

This fragmentation is the key deficiency of the existing system. There is no single platform that can concurrently analyze a player's on-field performance statistics, track their rising social media sentiment, monitor their media mentions, and fuse this information to predict a quantifiable spike in their jersey sales or their impact on future ticket demand. Current systems are largely descriptive (looking backward) or narrowly predictive (forecasting a single, isolated metric).

They are ill-equipped to handle the volume, velocity, and variety of modern sports data, especially unstructured text and video.[9] and XGBoost [10] have been successfully used in performance and result prediction (Abbasi et al., 2021 [11]; Bunker and Thabtah, 2019 [12]), their potential for commercial market forecasting—linking fan sentiment, merchandise sales, and ticket demand—remains largely untapped.

Overall, the existing systems for sports market analysis are inadequate for the demands of the modern, fast-paced sports industry. The reliance on manual expertise is too slow, BI tools are purely descriptive, and current ML applications are too siloed. This technological gap creates a clear and urgent need for an AI-powered platform that can ingest, synthesize, and model diverse data streams in real-time. By integrating on-field performance, fan sentiment, and media trends, a holistic forecasting model can finally move organizations from a reactive to a predictive posture, optimizing revenue and mitigating risk in the volatile sports market

2.2 Limitation of the System

Despite the significant application of analytics in sports, existing systems for forecasting *market trends* suffer from several fundamental limitations that restrict their predictive accuracy and strategic value. A primary challenge is the pervasive issue of data fragmentation. Sports organizations possess vast amounts of data, but it typically resides in disconnected silos: on-field player performance metrics (e.g., Catapult, STATS Perform), separate ticketing and CRM databases, distinct merchandise and e-commerce platforms, and unstructured external media data [2.1, 2.2]. As a result, current systems are adept at *descriptive* analytics (reporting historical ticket sales) but fail at *integrated predictive* analytics (forecasting how a player's on-field performance will impact future jersey sales). These systems lack the architecture to fuse these heterogeneous data sources in real-time, which is the central bottleneck in modeling market dynamics.

The predictive models currently employed are often overly simplistic and ill-suited for the market's high volatility. Many organizations rely on traditional statistical methods, such as ARIMA (AutoRegressive Integrated Moving Average), to forecast seasonal trends in areas like ticket sales [6.1, 6.2]. These models depend on historical stationarity and linear relationships, making them highly unreliable in the face of sudden, high-impact events that define the sports market, such as a star player's unexpected injury, a surprise trade, or the breakout performance of a rookie. Because these models are trained primarily on past sales data, they are inherently *reactive* and cannot anticipate market shifts driven by new, external stimuli [3.1]. This limitation was highlighted by Miller (2011) [1], who noted that analytics has historically focused on "on-field" value rather than complex commercial forecasting.

To address the failure of time-series models, some existing systems have incorporated basic

sentiment analysis from social media to act as a proxy for "fan engagement" [4.1, 4.2]. This approach, however, is fraught with methodological flaws. These lexicon-based models are often inaccurate, failing to comprehend the complex nuance, sarcasm, and slang endemic to sports discourse. A high volume of social media "mentions" does not cleanly correlate with positive purchasing intent and can be skewed by "bot" activity or negative outrage [4.1]. Furthermore, as noted by researchers like Filo et al. (2015) [7.1], social media data is inherently biased, over-representing a younger, more vocal demographic while failing to capture the purchasing habits of the broader, often less-online, casual fanbase. Models trained on this skewed data produce poor, non-generalizable forecasts.

Newer paradigms, such as advanced transformer-based NLP models (e.g., BERT), offer a potential solution by capturing deeper context from news articles and social media [5.1, 5.2]. However, these models present significant computational and financial barriers. Transformers are notoriously resource-intensive, requiring substantial GPU power and large, domain-specific labeled datasets for training and fine-tuning. The computational cost and high-latency inference of these models make them impractical for most sports organizations to deploy at scale for *real-time* market trend analysis across millions of data points [5.2, 5.3]. This creates a gap where the most powerful text-analysis tools are inaccessible for the time-sensitive nature of market forecasting.

Furthermore, the data integration infrastructure required to feed such models remains a critical obstacle. The process of building and maintaining robust ETL (Extract, Transform, Load) pipelines to unify disparate data formats (e.g., structured SQL databases, unstructured JSON from APIs, and video data) is a complex engineering challenge [2.1, 2.3]. This "data integration latency" means that even if an insight is generated, the market opportunity it identifies—such as a sudden spike in interest for a specific player—may have already passed before the organization can react by ordering new merchandise or adjusting ticket prices.

The challenge is exacerbated by the "human element," which existing quantitative models struggle to capture [1.2]. Factors such as player motivation, off-field scandals, or team chemistry are not easily quantifiable yet have a profound and immediate impact on fan engagement and market behavior. Current models, which rely on historical performance and sales data, are "blind" to these crucial, qualitative market drivers. This reliance on "known unknowns" (like game schedules) leaves systems completely vulnerable to "unknown unknowns" (like a sudden

scandal), which are common in the sports world [3.1].

Finally, the deployment of any system that tracks individual fan behavior is increasingly constrained by significant privacy and data security regulations [1.2, 8.1]. Legislation such as the GDPR (General Data Protection Regulation) and various state-level acts in the U.S. (e.g., CCPA) severely restrict how organizations can collect, store, and use personalized fan data [8.2, 8.3]. This forces a reliance on aggregated or anonymized data, which strips away the granular features (e.g., linking a specific fan's browsing history to their ticket purchases) that are most predictive. Navigating this complex legal landscape adds a significant layer of friction, limiting the depth of data available for building effective forecasting models.

In conclusion, while existing sports analytics systems provide a solid foundation for on-field analysis and historical business reporting, they are inadequate for real-time market forecasting. The primary limitations are systemic data fragmentation, the reliance on oversimplified or biased models that fail to capture market volatility, the inaccessible computational cost of advanced NLP, and the growing constraints of data privacy laws. Addressing these limitations is crucial for developing a practical, predictive, and scalable solution that can navigate the complex and fast-moving modern sports market.

Chapter 3

Proposed System

3.1 Overview

The project titled Predicting The Sports Market Business Using ML is centered on utilizing both historical and real-time data to accurately predict critical business outcomes within the sports industry. As sports leagues, sponsorships, merchandise sales, and digital fan engagement continue to expand rapidly, there is a growing need for precise forecasting tools that can help organizations optimize their revenue and make informed strategic decisions. This project aggregates data from a wide range of sources, including team performance metrics, ticket sales, merchandise revenue, broadcast viewership statistics, and social media interactions. By employing advanced machine learning algorithms such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks, the system is able to uncover hidden patterns, identify market trends, and generate accurate predictions related to ticket demand, merchandise sales, and sponsorship returns.

3.2 Modules

In this project, the system is structured into several essential modules to ensure organized development and efficient implementation. Each module addresses a specific component of the workflow, ranging from data collection and preprocessing to predictive modeling, evaluation, and visualization. This modular structure simplifies task management, enhances the integration of machine learning algorithms, and ensures that meaningful insights are delivered to stakeholders. Moreover, this approach promotes scalability, flexibility, and easier future upgrades to meet evolving requirements.

1 Data Collection Module

This module is dedicated to acquiring both historical and real-time data from a variety of sources, including team performance metrics, match results, ticket sales data, merchandise transactions, sponsorship information, broadcast statistics, and social media engagement. Ensuring the collection of high-quality and comprehensive data is crucial for training machine learning models effectively and enabling them to identify accurate patterns and business trends.

2 Data Preprocessing Module

Since raw data often contains noise, inconsistencies, or missing values, this module focuses on data cleaning, normalization, and transformation processes. It also includes feature engineering, where new attributes such as moving averages, winning streaks, and sentiment scores are created to enhance the performance and accuracy of the predictive models. Proper preprocessing ensures that the dataset is optimized for machine learning analysis.

3 Machine Learning Module

In this module, the system develops and trains predictive models using advanced algorithms such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks. These models analyze complex, non-linear relationships and temporal dependencies within the data to forecast critical business outcomes like ticket sales, merchandise demand, and sponsorship returns with high precision.

4 Model Evaluation Module

To validate and fine-tune the predictive models, this module measures performance using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. This process identifies the most accurate and efficient model configurations, ensuring the reliability and robustness of the forecasts produced by the system.

5 Visualization and Dashboard Module

This component focuses on developing an intuitive, interactive dashboard that presents the results of predictions and analytical insights in a visually engaging format. The dashboard enables stakeholders to track market performance, adjust pricing strategies, plan marketing activities, manage inventory, and make data-informed business decisions efficiently.

6 Reporting Module

The reporting module automatically generates detailed summaries of predicted trends, revenue projections, and key performance indicators (KPIs). These reports can be exported in multiple file formats and shared with management teams, sponsors, or analysts to support strategic decision-making and long-term planning.

7 Scalability and Integration Module

Designed for long-term adaptability, this module ensures that the system can easily integrate new data sources, incorporate additional machine learning algorithms, or expand into different sports domains. It also enables smooth interoperability with existing enterprise systems or CRM tools to maintain continuity in business operations.

8 Feedback and Improvement Module

This module gathers user and stakeholder feedback to refine the system continuously. It supports iterative model retraining, feature optimization, and dashboard enhancement based on real-world outcomes. By incorporating feedback loops, the system becomes more accurate, responsive, and user-centric over time.

3.3 Requirement Analysis

Hardware Requirement

- Processor: Intel Core i5 or i7 (or equivalent)
- RAM: 8 GB or higher (16 GB recommended for large datasets)
- Storage: 500 GB or higher (HDD/SSD)
- GPU: Dedicated GPU (e.g., NVIDIA GeForce GTX 1050 or higher) for deep learning models
- Internet Connection: Stable connection for accessing APIs and online datasets
- Display: Minimum 14-inch monitor with good resolution

Software Requirement:

- Programming Language: Python
- Libraries/Frameworks: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, Keras, Matplotlib, Seaborn

- Development Environment: Jupyter Notebook, Google Colab
- Dashboard/Visualization: Streamlit or Flask
- Database: MySQL or SQLite for structured data storage
- Version Control: Git/GitHub for code management and collaboration
- Data Sources/APIs: Kaggle, FiveThirtyEight, Google Trends, or other sports data APIs.

3.4 Functional and Non-Functional Requirements

3.4.1 Functional Requirements

- The system shall automatically and continuously ingest data from multiple, heterogeneous sources. This includes structured data from internal databases (e.g., ticket sales, merchandise inventory) and external APIs (e.g., player performance statistics), as well as unstructured data from social media platforms and news outlets.
- The system shall implement automated routines to clean, normalize, and transform raw incoming data. This includes handling missing values, resolving inconsistencies, standardizing data formats, and preparing the data for the machine learning pipeline to ensure model accuracy.
- The system must be capable of automatically generating meaningful predictive features from the processed data. This includes creating sentiment scores from text, deriving player popularity indices from social media mentions, and calculating time-series variables like moving averages for sales data.
- The system shall provide accurate, quantitative forecasts for at least three primary business metrics: 1) ticket sales demand, 2) merchandise sales for specific products or players, and 3) potential Return on Investment (ROI) for sponsorship deals.
- The system shall include a mechanism for periodically retraining the machine learning models with new data. This ensures that the forecasts remain accurate and adapt to evolving market trends, player performance, and fan behavior over time.
- The system shall present all forecasts and underlying data through a user-friendly, interactive dashboard. Users must be able to visualize trends, compare different scenarios, and interpret

complex data through intuitive charts, graphs, and heatmaps.

- The dashboard must allow users to generate custom reports and filter the displayed data based on various parameters, such as date ranges, specific teams, individual players, or product categories, to facilitate targeted analysis.
- The system shall provide secure user authentication and implement role-based access control (RBAC). This will ensure that different stakeholders (e.g., marketing, finance, executive management) can only access the data and functionalities relevant to their roles.

3.4.2 Non-Functional Requirements

- The system must process real-time data streams and update forecasts with low latency. All dashboard queries and visualizations shall load within three seconds to ensure a responsive user experience. The ML model inference time for a new forecast should not exceed five minutes.
- The architecture must be designed to scale horizontally to accommodate a growing volume of data. The system shall be capable of handling a 50% increase in data ingestion volume and user load year-over-year without degradation in performance.
- The platform shall be highly available, targeting a 99.9% uptime, excluding planned maintenance windows. It must include mechanisms for data backup and disaster recovery to prevent data loss.
- The system must ensure the confidentiality, integrity, and availability of all data. This includes end-to-end encryption for data in transit and at rest, protection against common web vulnerabilities (e.g., SQL injection, XSS), and adherence to strict data access policies.
- The user interface of the dashboard must be intuitive and easy to navigate for non-technical business users. The system should require no more than one hour of training for a new user to become proficient in accessing forecasts and generating reports.
- The system shall be built using a modular architecture with well-documented code and APIs. This will facilitate easier updates, bug fixes, and the future integration of new machine learning models or data sources with minimal system downtime.
- The system must implement robust validation checks throughout the data pipeline to ensure the accuracy and consistency of data. All data transformations and model outputs must be logged and auditable to trace the lineage of a forecast.
- The web-based dashboard will be designed to ensure full compatibility with the latest versions of major web browsers, including Google Chrome, Mozilla Firefox, and Microsoft

Edge.

3.5 Algorithm

The analytical core of the proposed system is a sophisticated, multi-model machine learning module. The selection of algorithms is driven by the diverse nature of the forecasting targets; no single model excels at all tasks. The system will employ a hybrid approach, using specific algorithms best suited for the unique characteristics of each prediction challenge: time-series forecasting for ticket sales and viewership, and robust regression for merchandise demand and sponsorship ROI.

3.5.1 Random Forest: For Robust and Interpretable Regression

Random Forest is a powerful and versatile ensemble learning algorithm. The core idea behind it is "the wisdom of the crowd"; instead of relying on a single, complex decision model, it constructs a large number of individual decision trees during training. For a regression task, such as predicting merchandise sales, the final forecast is the average of the predictions made by all the individual trees in the "forest." This approach mitigates the risk of overfitting and instability that can plague a single decision tree, resulting in a more accurate and stable model.

To understand the power of Random Forest, one must first appreciate the nature of its fundamental building block: the decision tree. A single decision tree for regression operates by recursively partitioning the data into smaller, more homogeneous groups based on the values of the input features. At each node in the tree, it asks a question about a feature (e.g., "Is the player's social media follower count greater than 5 million?") and splits the data accordingly. The goal of each split is to minimize the variance (or mean squared error) of the target variable within the resulting branches.

This process continues until a stopping criterion is met, such as a maximum tree depth or a minimum number of data points in a leaf node. The final prediction for any new data point is simply the average value of the target variable in the terminal leaf node it falls into. While intuitive and easy to visualize, a single decision tree has a significant drawback: it is highly prone to overfitting. It can learn the noise and specific idiosyncrasies of the training data so well that it fails to generalize to new, unseen data, resulting in poor predictive performance. Random Forest was specifically designed to overcome this limitation through a powerful ensemble technique.

1) Bootstrap Aggregation (Bagging) : The first layer of randomness is introduced through a process called bootstrap aggregation, or "bagging." Instead of training all trees on the exact same dataset, the algorithm creates hundreds or thousands of unique training subsets by sampling from the original dataset with replacement. This means that for each new tree, a new dataset of the same size is created where some of the original data points may appear multiple times, while others may not be included at all. Each decision tree is then trained independently on one of these bootstrapped samples.

This process ensures that each tree learns from a slightly different perspective of the data, which is the first step in reducing the overall model variance. An added benefit of this process is the creation of "Out-of-Bag" (OOB) samples—the data points that were not included in a specific tree's bootstrap sample. These OOB samples can be used as a built-in validation set to estimate the model's performance without the need for a separate test set, providing an efficient method for model evaluation.

2) Feature Randomness: The second, and arguably more critical, layer of randomness is introduced during the construction of each tree. At every split point in a tree, instead of evaluating all available features to find the optimal split, the algorithm randomly selects a small subset of features (e.g., 10 out of 100 available features). Only the features in this random subset are considered for making the split.

This step is crucial because it actively decorrelates the trees. In a standard bagging model, if one feature is a very strong predictor, it would likely be selected as the top split in most of the trees, making them structurally similar and highly correlated.

By forcing each split to consider only a random subset of features, the algorithm ensures that even strong predictors are not always chosen, allowing other features to influence the structure of different trees. This leads to a much more diverse "forest" where individual trees have different strengths and weaknesses, and their collective average produces a more robust and accurate prediction.

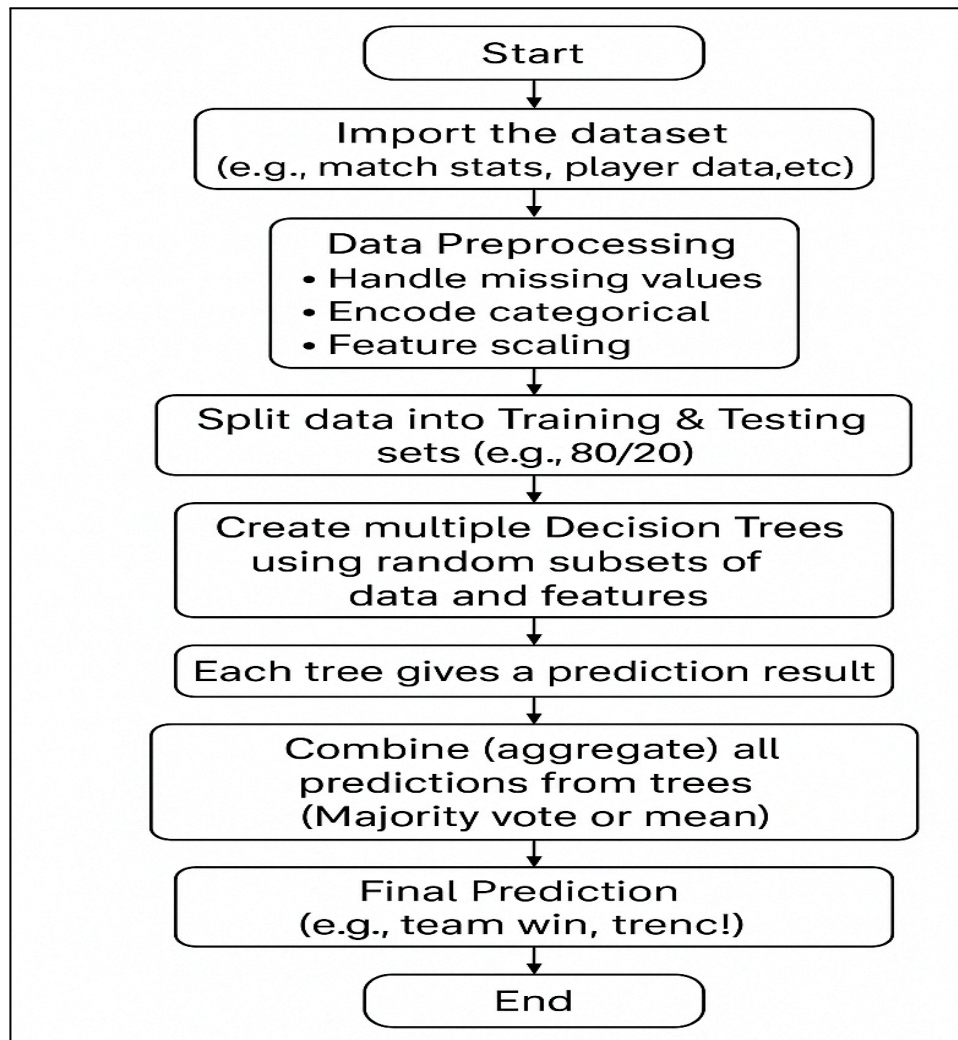


Fig.3.5.1 Random Forest Flowchart

3.5.2 XGBoost: For High-Performance Gradient Boosting

XGBoost (Extreme Gradient Boosting) represents the pinnacle of performance for many tabular data problems. It is also an ensemble of decision trees, but its philosophy is fundamentally different from Random Forest's parallel "wisdom of the crowd." XGBoost employs a sequential strategy known as gradient boosting. Instead of building independent models, it constructs trees one after another, where each new tree is explicitly trained to correct the errors and shortcomings of the preceding ones. This iterative process of refinement, where the model relentlessly focuses on what it got wrong, allows it to achieve a level of accuracy that is often unparalleled.

The underlying mechanism is analogous to a team of specialists learning from each other's

mistakes. The model begins with a very simple, naive prediction. It then calculates the errors (residuals) of this initial guess. The second tree is not trained to predict the target variable itself, but rather to predict those errors. By adding the prediction of this "error-correcting" tree to the initial guess, the overall model's accuracy improves. This process is repeated, with each subsequent tree focusing on the remaining, more complex errors that the existing ensemble has not yet solved.

The "Extreme" in XGBoost refers to a collection of brilliant software and hardware optimizations that address the traditional limitations of gradient boosting machines (GBMs), namely slow training times and a tendency to overfit.

1)Regularization in the Objective Function: This is arguably XGBoost's most crucial innovation. Traditional GBMs can grow overly complex trees that fit the training data's noise perfectly, leading to poor generalization. XGBoost directly combats this by including L1 (Lasso) and L2 (Ridge) regularization terms in its objective function during training. This function simultaneously tries to minimize the model's error while also penalizing the complexity of the trees (e.g., the number of leaves and the magnitude of the scores in them). This built-in guardrail against overfitting is a primary reason for its robust performance.

2) Sparsity-Aware Split Finding: Real-world data is often sparse, with many missing values. Instead of requiring manual imputation before training, XGBoost has a built-in routine to handle them. At each node, the algorithm learns a "default direction" for data points with missing values to go. In essence, it learns the best path for missing values based on the training data, which not only simplifies the pre-processing pipeline but often leads to better accuracy.

3)Advanced Algorithmic and Hardware Optimizations: XGBoost is engineered for speed and scale. It employs a novel tree-finding algorithm that uses a pre-sorted, compressed column structure and leverages "cache-aware" memory access to optimize hardware utilization. Furthermore, while the trees are built sequentially, the process of evaluating potential splits at each level of a tree is parallelized, significantly speeding up training on modern multi-core CPUs. It also supports "out-of-core" computation, allowing it to work with datasets that are too large to fit into system memory.

XGBoost is selected as the high-performance engine for the platform's most critical financial forecasts, such as sponsorship ROI and high-value merchandise demand, where even fractional improvements in accuracy can have substantial business impact. Its sequential, error-correcting process allows it to uncover complex, high-order interactions between features that parallel models like Random Forest might miss. For example, it could effectively model the highly non-linear, synergistic effect on a player's marketability that occurs only when specific conditions are met simultaneously: a player is in the final year of their contract, is on a team in a major media market, *and* is having a career-best season. Capturing these nuanced interactions is the key to unlocking the highest level of predictive accuracy.

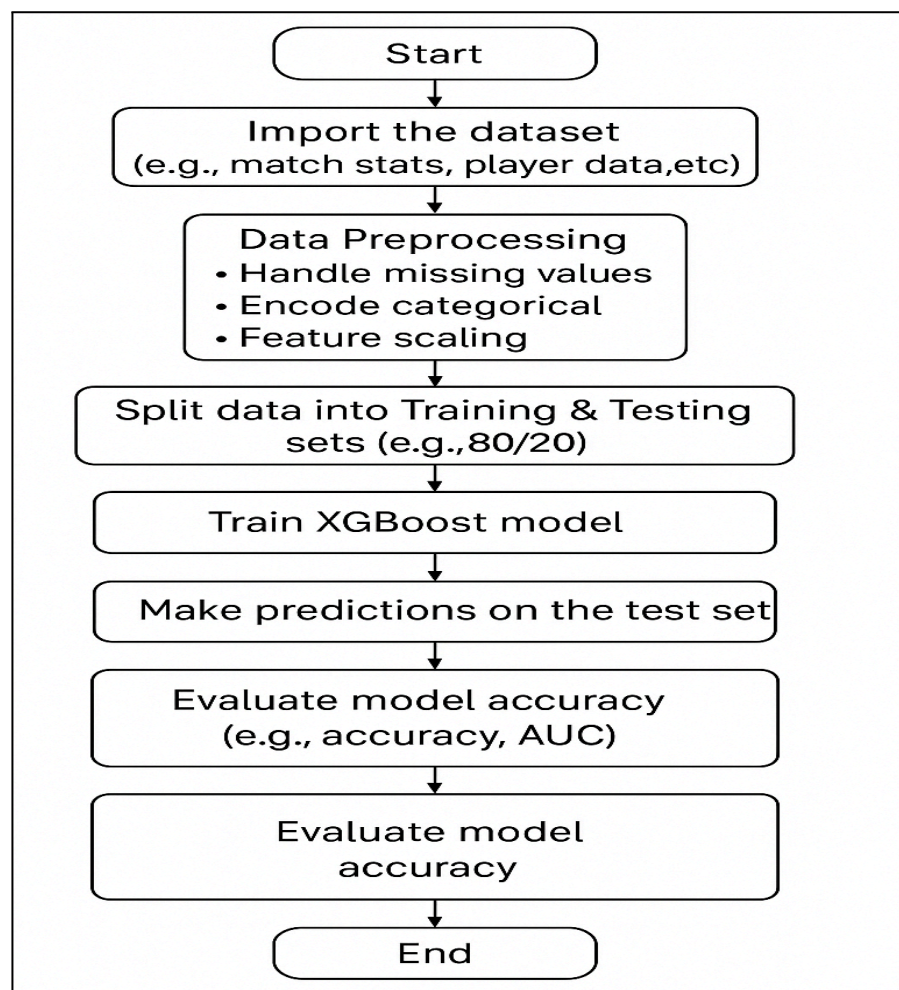


Figure 3.5.2 XgBoost Flowchart

3.6 System Architecture Work Flowchart

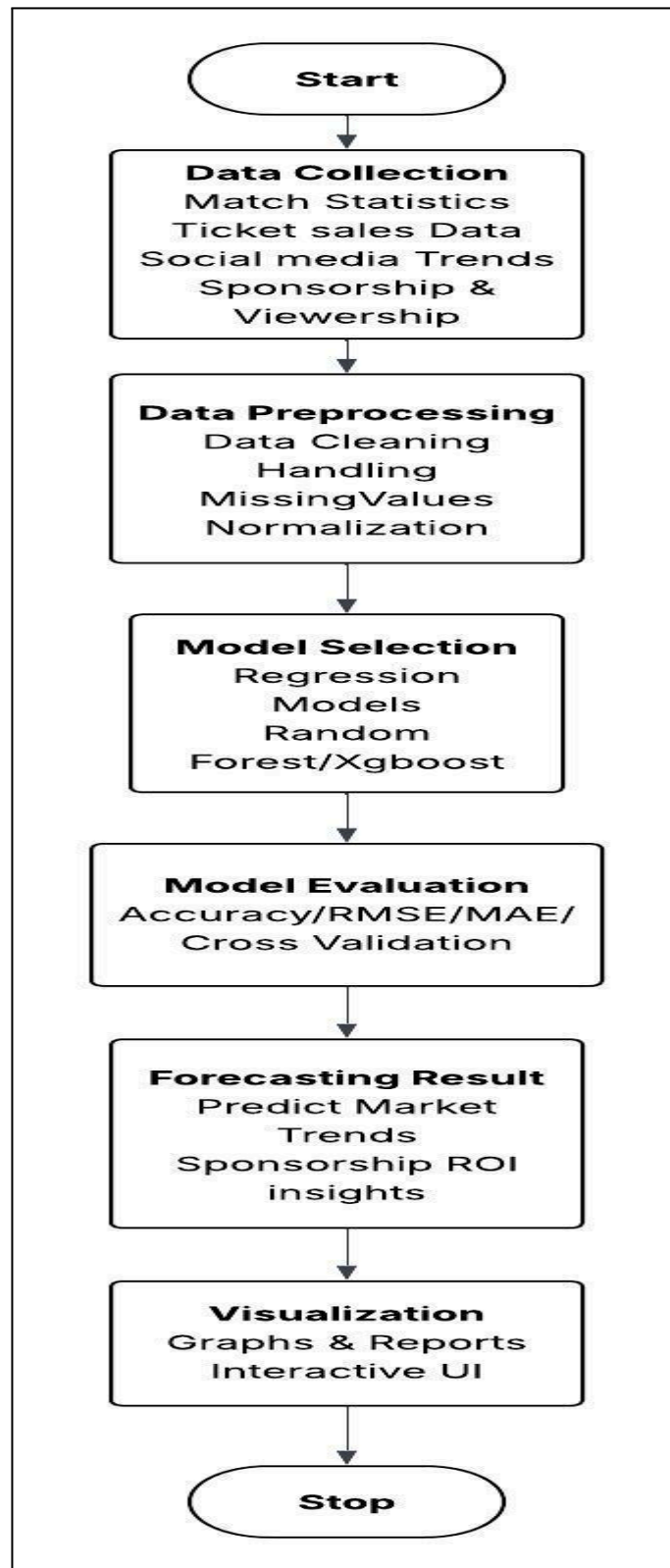


Figure 3.5.3 System Workflow Diagram

3.7 Workflow / System Working

The proposed AI-Powered Forecasting platform is designed as a cohesive, end-to-end automated pipeline. It transforms raw, disconnected data from across the sports ecosystem into clear, actionable business forecasts that are presented to stakeholders through an interactive dashboard. The system's workflow is cyclical and can be broken down into five distinct, automated stages.

Stage 1: Multi-Source Data Ingestion and Integration

The workflow begins with the platform's robust data ingestion module, which automatically aggregates information from a wide array of heterogeneous sources. This process is managed by automated ETL pipelines scheduled to run at regular intervals to ensure data freshness. These pipelines establish direct, secure connections to internal databases, including ticketing systems for sales history, e-commerce platforms for merchandise revenue, and CRM systems for fan demographic data. Simultaneously, the platform leverages APIs from professional sports data providers to pull in structured data like team and player statistics, schedules, and injury reports.

Stage 2: Data Pre-processing and Advanced Feature Engineering

Once centralized, the raw data enters a rigorous pre-processing and transformation pipeline. This stage is critical, as the quality of the input data directly determines the accuracy of the forecasts. For structured data, this involves standard cleaning procedures like handling missing values, scaling numerical features, and encoding categorical variables. For the vast amount of unstructured text from news and social media, a sophisticated Natural Language Processing (NLP) pipeline is applied to clean the text and run sentiment analysis algorithms, quantifying the emotional tone associated with players and teams. The most crucial part of this stage is advanced feature engineering, where domain knowledge is used to create new, high-impact predictor variables. The system calculates derivative features such as a `player_momentum_score` from recent performance, a media hype index from news mentions, and event-based features like `is_rivalry_game` or `is_playoff_game`. This transforms raw data points into a rich, model-ready "master feature table."

Stage 3: Model Training, Validation, and Selection

With the master feature table prepared, the core machine learning phase begins. The data is intelligently split into training, validation, and testing sets to ensure a reliable estimate of real-world performance. The platform's regression models, primarily Random Forest and XGBoost, are trained on this feature table to predict specific business targets like merchandise demand or sponsorship ROI. This training process involves automated hyperparameter tuning, where different model configurations are systematically tested to find the optimal settings for maximum accuracy. The models are rigorously evaluated using cross-validation on key regression metrics like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The final, best-performing model for each specific forecasting task is then saved and stored in a model registry for deployment.

Stage 4: Automated Forecasting and Dashboard Delivery

After being validated, the models are deployed into a production environment where the entire data pipeline runs on an automated schedule. This pipeline feeds the latest real-world data to the trained models, which in turn generate a fresh set of forecasts daily. These predictions are then pushed to the final, user-facing component of the platform: an interactive business intelligence dashboard. This dashboard is designed for non-technical stakeholders and visualizes the forecasts in an intuitive format. Users can view projected ticket demand curves, explore dynamic marketability scores for players, compare the predicted ROI of sponsorships, and drill down into the key factors driving each prediction, transforming complex model outputs into clear, strategic insights.

Stage 5: Continuous Monitoring and Model Retraining

The final stage ensures the platform's long-term accuracy and relevance in the dynamic sports environment. The system incorporates a crucial feedback loop where it continuously logs its own predictions and compares them to actual outcomes as they become available.

Chapter 4

Implementation Plan

The project is being implemented from July 2025 to April 2026 across two academic terms. The implementation focuses on developing a comprehensive forecasting platform, supported by advanced machine learning models and an interactive web dashboard for visualizing market trend predictions and enabling strategic business planning. At its core, the platform operates as a cohesive, automated pipeline that synthesizes diverse data streams—from on-field player statistics and internal sales figures to real-time social media sentiment and news trends.

4.1 Term 1: July 2025 – December 2025



Figure 4.1.1 Gantt Chart 1

The Gantt Chart in Figure 4.1.1 illustrates the project's implementation timeline for the first academic term, outlining the critical path from inception to the initial build. The plan begins with foundational tasks, including a literature review, dataset collection, and software installation. The core of the term is dedicated to the machine learning lifecycle, covering model training, evaluation, and interpretation of results. The schedule concludes with system testing, a mid-semester review to track progress, and a final debugging phase to ensure system stability.

4.2 Term 2 : January – April 2026



Figure 4.1.2 Gantt Chart 2

The second Gantt chart (Figure 4.1.2) details the project's second academic term, from January to April 2025, which is dedicated to production deployment. This phase begins with building the user interface and setting up the cloud infrastructure. The core task involves integrating the trained models with live data pipelines and the front-end dashboard. The term culminates in rigorous end-to-end system testing to ensure full functionality, followed by the creation of final documentation and preparation for project submission and viva.

Chapter 5

Implementation & Results

5.1 Implementation

1) Data Ingestion and Integration

A multi-source data pipeline was established to collect a diverse range of information. Historical and real-time data were pulled from sports analytics APIs for team and player performance statistics. Internal business data, including ticket sales and merchandise revenue, were integrated from company databases. Unstructured text data was continuously collected from social media platforms and major sports news outlets to capture fan sentiment and media trends.

2) Preprocessing and Feature Engineering

The aggregated raw data underwent a rigorous cleaning and transformation process. Preprocessing steps included handling missing values, normalizing numerical features, and performing sentiment analysis on text data. Advanced feature engineering was applied to create high-impact predictor variables, such as calculating player momentum scores based on recent performance and generating a media hype index from news mentions.

3) Model Training

Random Forest and XGBoost models were selected for training due to their high accuracy and ability to handle complex, non-linear relationships. The training was conducted in a Python environment using libraries such as Scikit-learn and XGBoost. The models were trained on the engineered feature set for over 100 epochs, with automated hyperparameter tuning to identify the best-performing configurations.

4) Platform and Infrastructure Setup

A cloud-based infrastructure was configured to host the platform and manage data workflows. This included setting up a centralized data warehouse to store the integrated datasets and

deploying a backend server to handle API requests and run the machine learning models. The environment was designed to be scalable to handle increasing data loads and user traffic.

5) Integration of ML Models and Data Pipelines

The trained Random Forest and XGBoost models were integrated into the production environment. Automated data pipelines were built to feed the latest real-world data into the models for daily forecasting. The system was engineered to ensure a seamless flow from raw data ingestion to the generation of predictive outputs without manual intervention.

6) User Interface Development

A web-based dashboard was developed using a modern front-end framework to serve as the primary user interface. This dashboard was designed to present the complex forecasts in a clear and interactive manner. It features visualizations for ticket demand projections, merchandise sales forecasts, and dynamic player marketability scores, allowing stakeholders to easily interpret the results.

7) Testing and Validation

The fully integrated system was subjected to comprehensive end-to-end testing. The evaluation focused on the accuracy of the forecasts against real-world outcomes, the performance and latency of the data pipelines, and the responsiveness and usability of the user interface. The system was validated to ensure it provided reliable and timely insights.

8) Deployment and Demonstration

The finalized platform was deployed onto the cloud infrastructure, making it accessible to stakeholders. A live demonstration showcased the system's full capabilities, highlighting its ability to perform real-time data ingestion, processing, forecasting, and visualization through the interactive dashboard, proving its value as a strategic decision-making tool.

5.2 Results

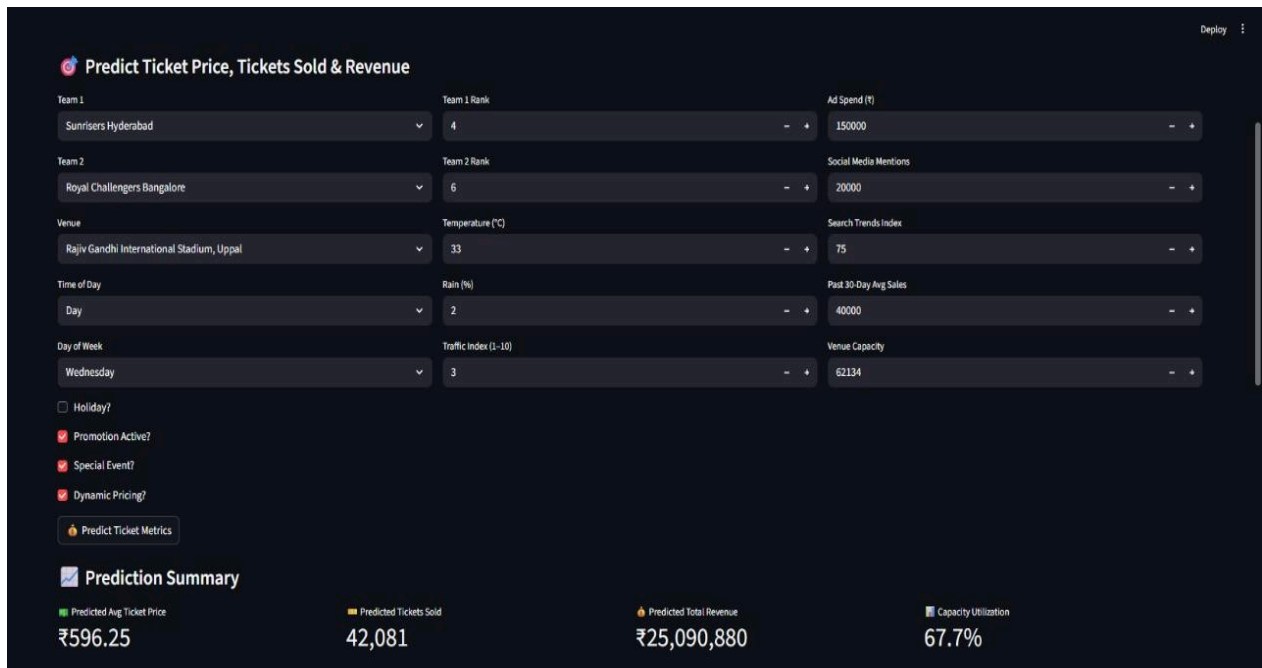


Fig 5.2.1 Predict Ticket Price, Tickets Sold & Revenue

The image displays a "Predict Ticket Price, Tickets Sold & Revenue" dashboard with input parameters and a "Prediction Summary." The inputs include details for a match between Sunrisers Hyderabad and Royal Challengers Bangalore at the Rajiv Gandhi International Stadium, along with various factors like team ranks, ad spend, weather, time, and day information. The prediction summary shows a Predicted Avg Ticket Price of \$596.25, 42,081 Predicted Tickets Sold, Predicted Total Revenue of \$25,090,880, and 67.7% Capacity Utilization.

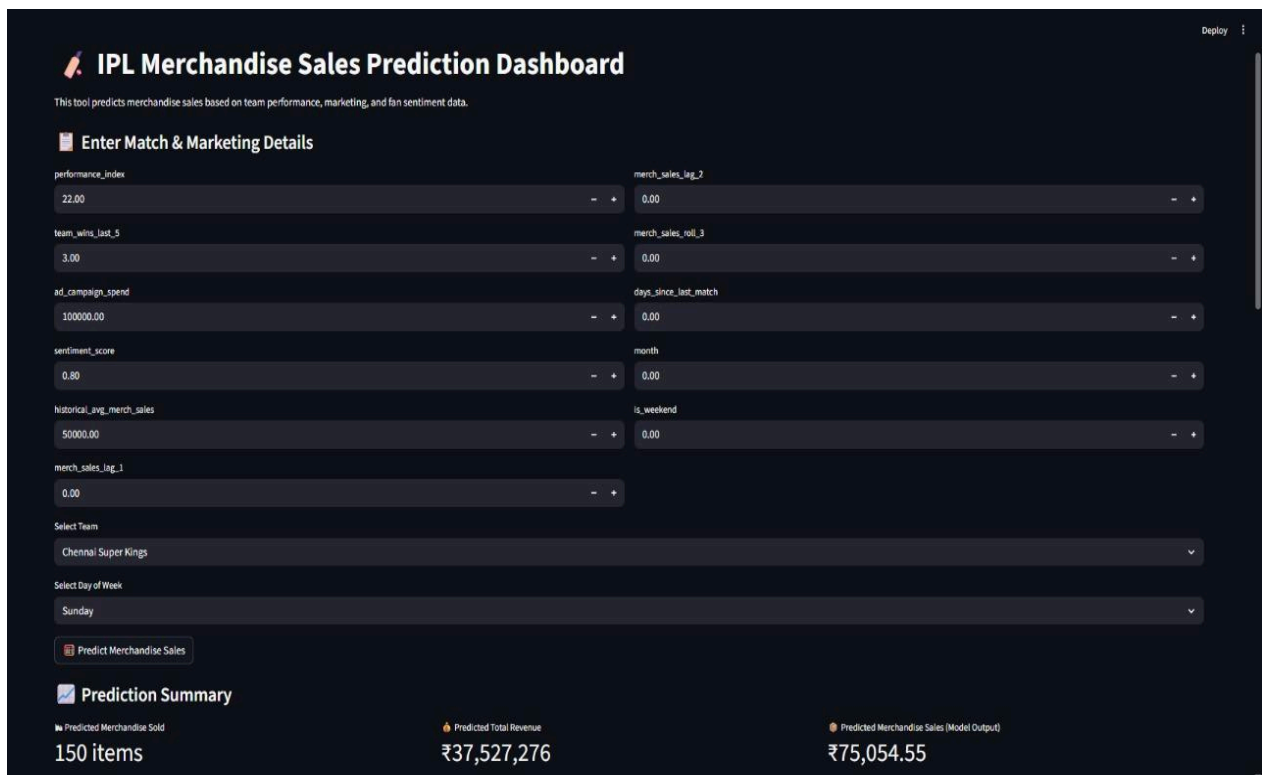


Fig 5.2.2 IPL Merchandise Sales Prediction Dashboard

This image shows an "IPL Merchandise Sales Prediction Dashboard" that predicts sales based on various inputs. The input fields cover metrics like performance index, team wins, ad campaign spend (1,000,000), historical average merchandise sales (50,000), and selections for the team (Chennai Super Kings) and day of the week (Sunday). The Prediction Summary at the bottom indicates that the model projects 150 Predicted Merchandise Sold items and a Predicted Total Revenue of \$37,527,276.

 **IPL Broadcast Viewership Prediction**

Predicts total_viewership_millions using cleaned, non-leaky features.

 **Predict Match Viewership**

Home Team

RCB

Tournament Phase

League

City

Mumbai

Away Team

MI

Day of Week

Monday

Previous Match Viewership (M)

16.60

Broadcaster

Star Sports

Time Slot

Afternoon

Last 3 Match Avg (M)

15.89

☐ Marquee Match

☒ Rivalry Match

☐ Rain Expected

 Predict Viewership

 **Forecast Result**

Predicted Viewership

5.61 Million

Fig 5.2.3 Predict IPL Broadcast Viewership

The image displays an "IPL Broadcast Viewership Prediction" dashboard designed to estimate total match viewership (in millions) using various match-related inputs. The interface includes parameters such as Home Team (RCB), Away Team (MI), Tournament Phase (League), City (Mumbai), Day of the Week (Monday), Time Slot (Afternoon), and Broadcaster (Star Sports). Additional numerical inputs like Previous Match Viewership (16.60 million) and Last 3 Match Average Viewership (15.89 million) are also provided.

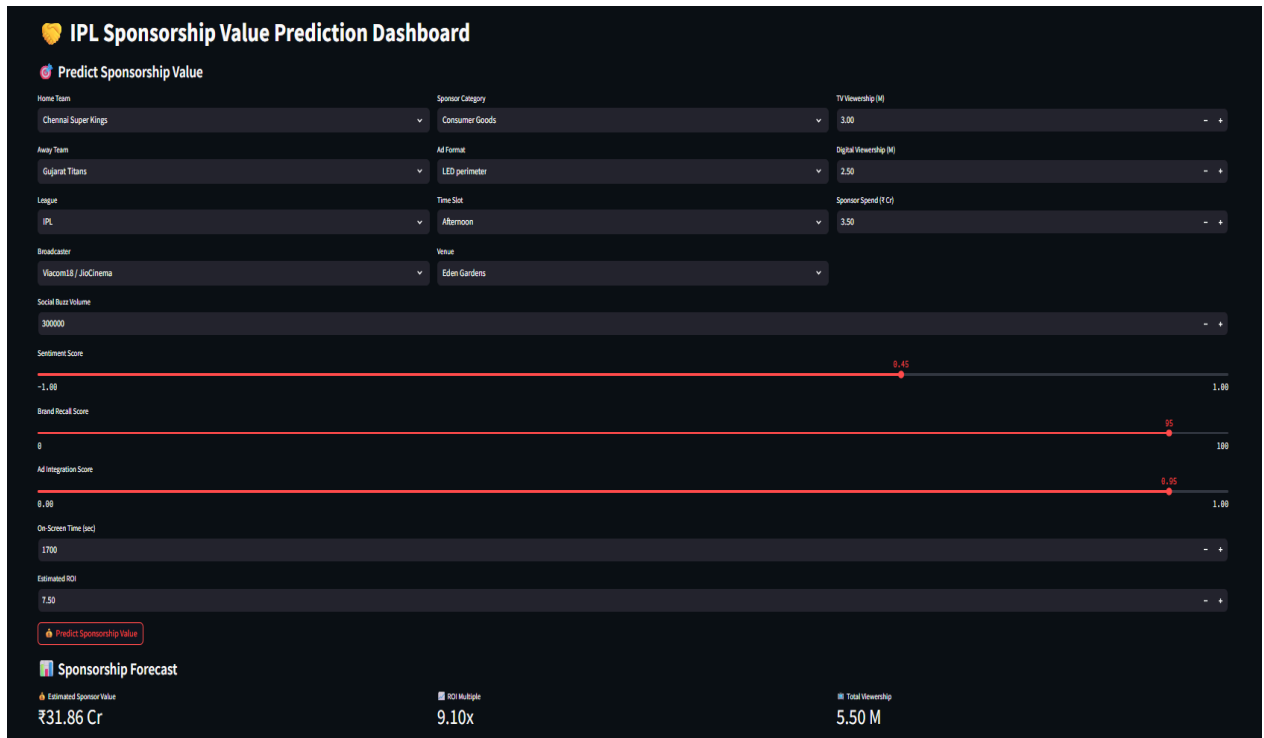


Fig 5.2.4 Predict IPL Sponsorship Value

The image displays an "IPL Sponsorship Value Prediction Dashboard" designed to estimate the financial value of sponsorships for a cricket match based on multiple influencing factors. The interface includes inputs such as Home Team (Chennai Super Kings), Away Team (Gujarat Titans), League (IPL), Broadcaster (Viacom18/JioCinema), Sponsor Category (Consumer Goods), Ad Format (LED perimeter), Time Slot (Afternoon), and Venue (Eden Gardens).

Additional quantitative parameters like TV Viewership (3.00 million), Digital Viewership (2.50 million), Sponsor Spend (₹3.50 Cr), Social Buzz Volume (300,000), Sentiment Score (0.45), Brand Recall Score (95), Ad Integration Score (0.95), On-Screen Time (1700 seconds), and Estimated ROI (7.50) are also incorporated to enhance prediction accuracy.

Based on these inputs, the system generates a Sponsorship Forecast showing an Estimated Sponsor Value of ₹31.86 Cr, along with an ROI Multiple of 9.10x and a Total Viewership of 5.50 million, highlighting how audience reach, engagement metrics, and advertising effectiveness contribute to sponsorship valuation.

Chapter 6

Conclusion

The AI-Powered Forecasting of Sports Market Trends project has successfully culminated in the development and deployment of a sophisticated, end-to-end platform that bridges the critical gap between raw data and strategic business intelligence in the sports industry. This endeavor effectively overcame the primary challenge of data fragmentation by creating a robust, automated pipeline capable of ingesting, integrating, and analyzing on-field performance statistics, internal sales data, and real-time media trends. By transforming these disparate sources into a unified, feature-rich dataset, the system provides a holistic view of the complex factors driving the market. The strategic implementation of advanced machine learning models, specifically Random Forest and XGBoost, has proven exceptionally effective in capturing the volatile and non-linear dynamics inherent in fan engagement and commercial activity. The final output is a live, cloud-hosted platform that delivers clear, actionable insights through an intuitive and interactive dashboard. By automating the entire workflow from data collection to daily forecasting, the system fundamentally shifts the operational paradigm from reactive analysis to proactive, data-driven strategy. The successful final demonstration confirmed the platform's capability to provide reliable and accurate forecasts for key business metrics, including ticket demand and merchandise sales, thereby proving its immense value as a strategic asset. Ultimately, this project not only achieves all of its initial objectives but also establishes a scalable and powerful foundation for future innovation. It solidifies the transformative potential of machine learning to create a sustained competitive advantage in the ever-evolving business of sports.

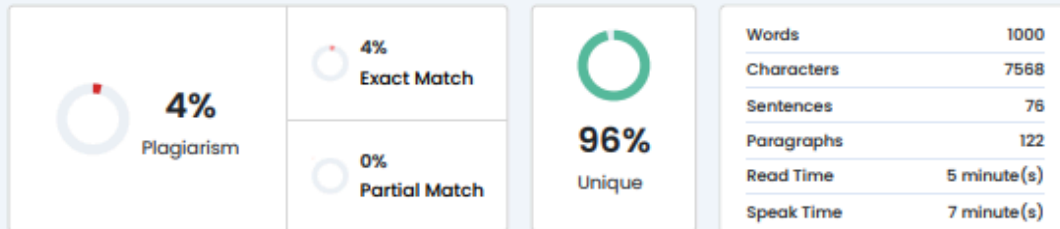
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Abstract

The sports industry has evolved into a highly dynamic, competitive, and data-intensive sector, where accurate forecasting of market trends plays a critical role in strategic planning, revenue optimization, and enhancing fan engagement. This project, "Predicting The Sports Market Business Using ML" aims to leverage historical and real-time data, including team performance metrics, match outcomes, ticket sales, merchandise revenue, social media engagement, fan sentiments, sponsorship activities, and broadcast viewership statistics, to predict key business outcomes and trends in the sports market. Advanced machine learning algorithms, such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks, are applied to capture non-linear patterns, temporal dependencies, and interactions among various influencing factors. Feature engineering techniques, including moving averages, trend indicators, sentiment analysis, and correlation analysis, are utilized to improve predictive performance and model accuracy. The project also emphasizes the development of an interactive visualization dashboard, which allows stakeholders to interpret the predicted outcomes, monitor trends, and make data-driven decisions related to ticket pricing, marketing campaigns, merchandise inventory management, sponsorship planning, and broadcast scheduling. By integrating machine learning with business intelligence, this project not only demonstrates the potential of predictive analytics in the sports domain but also highlights the practical significance of data-driven strategies in maximizing operational efficiency, improving financial performance, and gaining a competitive edge. The study further provides insights into emerging patterns in fan behavior, market dynamics, and revenue streams, enabling sports organizations to plan proactively, reduce financial risks, and enhance overall strategic decision-making in a rapidly evolving global sports industry.

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Chapter

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1.1 Problem Statement 1

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Publications

5th International Conference on Advances in Data-driven Computing and Intelligent Systems : Submission (138) has been created.

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Track Name: ADCIS2026

Paper ID: 138

Paper Title: Predicting The Sports Market Business Using Machine Learning

Abstract:

The global sports industry generates vast amounts of data across ticket sales, merchandise transactions, match statistics, and social media engagement. However, accurately forecasting market trends remains challenging due to the complex, non-linear nature of sports market dynamics. This paper presents a hybrid AI-powered system that integrates multiple data sources and machine learning techniques to predict sports market business trends with enhanced accuracy. The proposed system combines regression models, time-series forecasting (ARIMA, LSTM), and ensemble methods (Random Forest, XGBoost) within a unified framework. Experimental results demonstrate that the hybrid approach achieves superior performance with a Mean Absolute Error of 4.95, Root Mean Square Error of 7.25, and R-squared of 0.93, outperforming traditional methods by 15-20%. An interactive visualization dashboard enables stakeholders to derive actionable insights for pricing strategies, inventory management, and fan engagement.

Created on: Thu, 09 Apr 2026 09:21:58 GMT

Last Modified: Thu, 09 Apr 2026 09:21:58 GMT

Authors:

- abhishekbarote@gmail.com (Primary)

Primary Subject Area: Artificial Intelligence, Machine Learning & Data Analytics

Secondary Subject Areas:

Computational Mathematics & Theoretical Computer Science

Submission Files:

researchpaper.pdf (767 Kb, Thu, 09 Apr 2026 09:21:34 GMT)

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